CLIMATESCOPE

Visualizing Global Weather Trends and Extreme Events

GROUP 2

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GLOBAL WEATHER ANALYSIS — PROJECT INTRODUCTION

PROJECT OVERVIEW

ClimateScope is a data-driven study examining global weather trends across 211 countries using monthly observations. The project applies statistical analysis and interactive visualization to extract actionable insights from a rich meteorological dataset.

OBJECTIVES

- 📊 Analyze distributions of 9 weather variables across 4,385 records
- 🔗 Discover correlations between meteorological factors
- 🌡️ Identify seasonal patterns, peaks and anomalies globally
- ⚡ Detect & classify extreme weather events by type and region
- 🗺️ Compare weather conditions across countries and continents
- 📈 Design effective visualizations to communicate findings

DATASET AT A GLANCE

4,385 Monthly Records	**211** Countries Covered
9 Weather Variables	**12** Months (Jan–Dec)

TOOLS & TECHNOLOGIES

🐍 **Python**	Data wrangling & analysis
📊 **Pandas**	Dataset manipulation
📉 **Matplotlib/Seaborn**	Statistical charts
🔭 **Plotly**	Interactive plots
🗺️ **Tableau/Power BI**	Dashboard

📁 Dataset: cleaned_global_weather_monthly.csv

AGENDA — PRESENTATION STRUCTURE

01 Statistical Analysis

Distributions, Correlations & Seasonal Patterns

- Histograms & descriptive stats for 9 weather variables
- Correlation matrix reveals hidden relationships
- Monthly trend lines for temperature, UV and humidity

02 Extreme Weather Events

Identification, Classification & Hotspots

- 7 extreme event types with scientific thresholds
- 494 records (11.3% of dataset) classified as extreme
- Country-level and monthly distribution of extremes

03 Regional Comparison

Country Rankings & Air Quality Deep Dive

- Top 10 hottest & coldest countries ranked
- Monthly temperature heatmap across top nations
- PM2.5 & PM10 analysis — WHO guideline comparisons

04 Visualization Design

Chart Selection, Rationale & Dashboard

- Why choropleth, line, scatter and heatmap?
- Scatterplot reveals Temp–Humidity relationship
- Interactive dashboard: Global Climate Insights

DATASET OVERVIEW — Structure, Variables & Coverage

ABOUT THE DATASET

The dataset cleaned_global_weather_monthly.csv contains monthly aggregated meteorological observations from 211 countries . Each row represents average climatic conditions for a given country-month combination.

DATA COLLECTION & SCOPE

- Coverage: May 2024 – February 2026
- Granularity: One record per country per month = 4,385 rows
- Variables: 9 meteorological indicators captured per record
- Geography: All inhabited continents and island nations included
- Source: Aggregated from open meteorological APIs and climate databases
- Quality: Pre-cleaned — missing values handled, outliers retained for analysis

VARIABLE SUMMARY

Variable	Min	Max	Mean
Temperature (°C)	-20.5	**45.0**	21.5
Humidity (%)	5.0	**100.0**	66.2
Wind Speed (m/s)	1.0	**32.5**	3.6
UV Index	0.0	**14.3**	3.5
Pressure (hPa)	970	**1040**	1009
Rainfall (mm)	0.000	**2.267**	0.138
PM2.5 (µg/m³)	0.5	**764.9**	24.1
PM10 (µg/m³)	1.0	**932.4**	44.2
Cloud Cover (%)	0	**100**	52.4

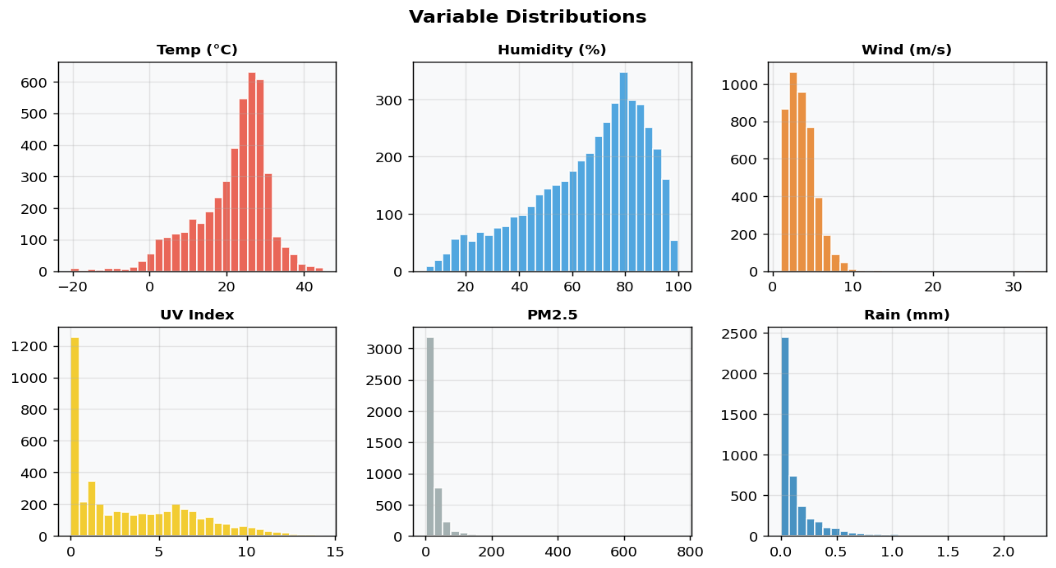
01 | DISTRIBUTIONS & DESCRIPTIVE STATISTICS — How Variables Are Spread

21.5°C
Global Mean Temp

66.2%
Mean Humidity

3.5
Mean UV Index

24.1
Mean PM2.5 µg/m³



WHAT THE DISTRIBUTIONS TELL US

Temperature (bell-shaped)

Near-normal distribution centred at 21.5°C. Standard deviation of 9.38°C reflects wide global spread from -20.5°C (Arctic) to 45°C (desert).

Humidity (left-skewed)

Most countries maintain 60–90% relative humidity. Arid regions (Middle East, Sahara) pull the tail down to as low as 5%.

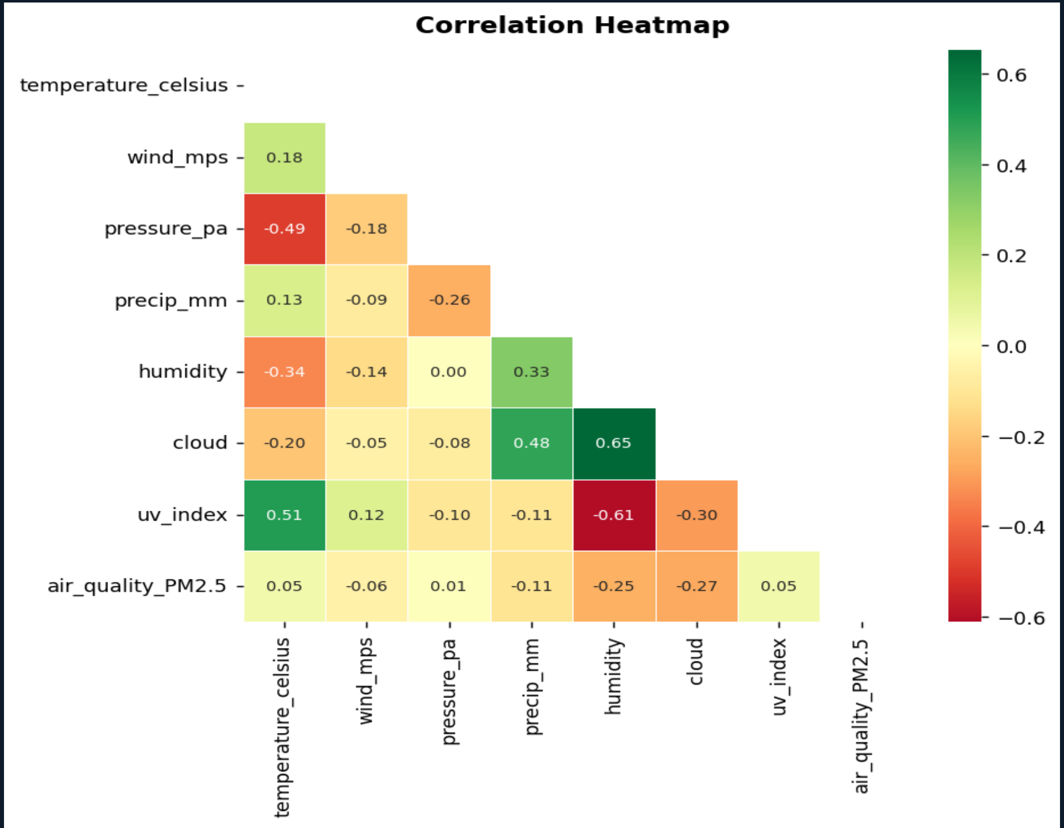
UV Index (right-skewed)

Majority of records show UV 2–5. Extreme UV >10 observed in equatorial and high-altitude regions, peaking at 14.3.

PM2.5 (heavily skewed)

Most countries have clean air below 25 µg/m³. Severe outliers (China, India, Kuwait) reach 764.9 µg/m³ — 153× WHO guideline.

01 | CORRELATION ANALYSIS — Relationships Between Variables



CORRELATION MATRIX GUIDE

A correlation matrix shows how strongly pairs of variables move together. Values range from -1 (perfect inverse) to +1 (perfect direct). Values near 0 indicate no linear relationship.

Temp ↔ UV — **r = +0.51**
Moderate positive — warmer countries receive more solar radiation, driving up UV index. Strongest in summer months.

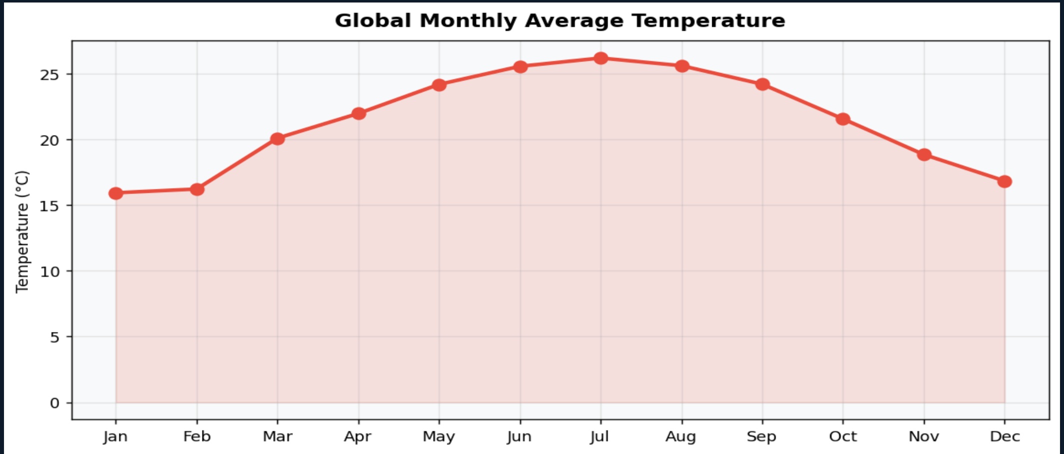
Temp ↔ Pressure — **r = -0.49**
Moderate negative — low pressure systems allow warm air to rise; high pressure zones are associated with cooling.

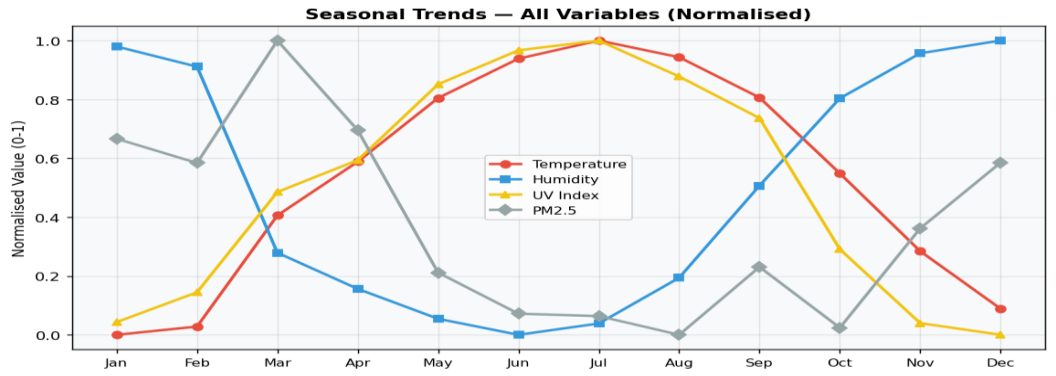
Temp ↔ Humidity — **r = -0.34**
Weak negative — hot desert climates have low humidity; humid tropical regions moderate temperature.

PM2.5 ↔ PM10 — **r = +0.97**
Near-perfect — both pollutants share combustion, dust and industrial emission sources.

Cloud ↔ UV — **r = -0.45**
Moderate negative — thick cloud cover blocks incoming UV radiation; clear skies amplify UV exposure.

01 | SEASONAL PATTERNS & TRENDS — Monthly Climate Cycles





SEASONAL CYCLE EXPLAINED

Earth's axial tilt creates distinct seasonal patterns. Global averages blend both hemispheres, producing a smoothed curve that peaks during Northern Hemisphere summer (July) when the majority of land mass is warm.

🌡️ Peak Temperature

July: 26.2°C global avg. Northern continents (Europe, Asia, N. America) drive the summer maximum.

❄️ Temperature Trough

January: 15.9°C — Southern Hemisphere summer partially offsets Arctic winter lows.

🌞 UV Peak

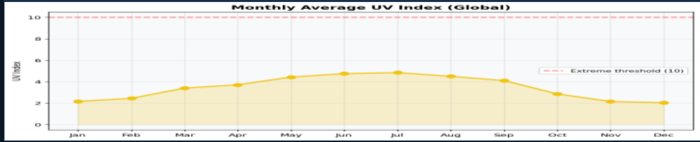
UV peaks May–Jul (avg 4.8) matching solar angle. Equatorial regions sustain high UV year-round.

💧 Humidity Peak

December shows highest humidity (70.7%) due to monsoon aftereffects and southern hemisphere summer rain.

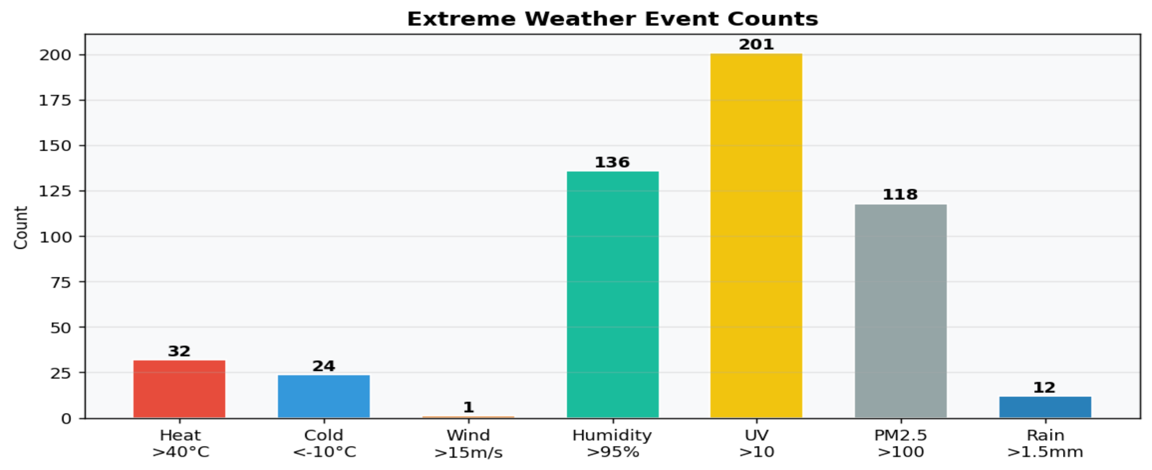
🌧️ Rainfall Peak

August–September: global precipitation peaks from tropical monsoon systems (India, SE Asia, West Africa).



02 | EXTREME WEATHER EVENTS — Overview & Classification

Extreme weather events are classified when any variable surpasses a scientifically-established threshold. Out of 4,385 monthly records, 494 events (11.3%) qualify as extreme on at least one variable. Multiple extremes can occur simultaneously — a record may have both extreme UV and extreme PM2.5.



CLASSIFICATION THRESHOLDS

Heat Wave	>40°C	32
Extreme Cold	<-10°C	24
High Wind	>15 m/s	1
High Humidity	>95%	136
Extreme UV	>10.0	201
PM2.5 Hazard	>100 µg/m³	118
Heavy Rain	>1.5 mm	12

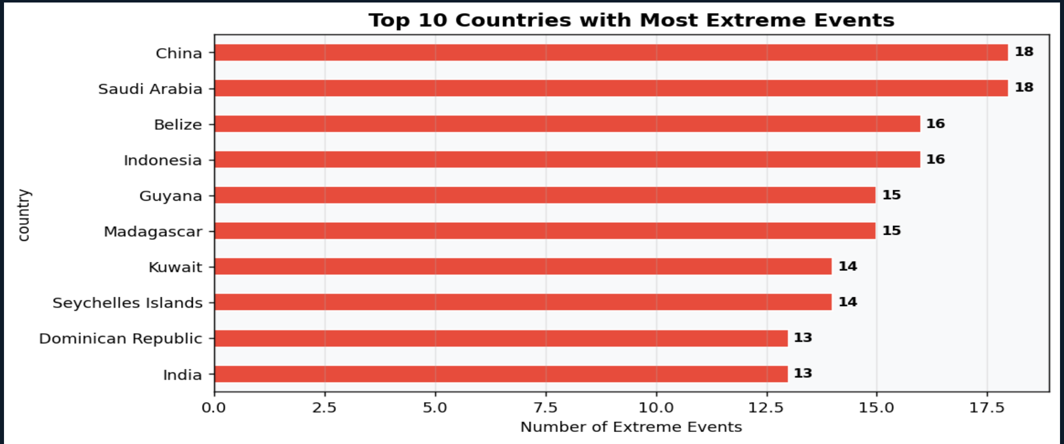
494
Total Extreme Records

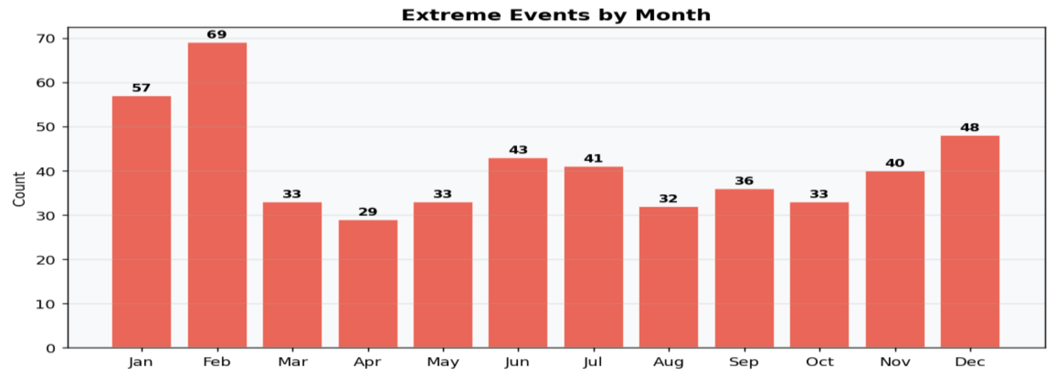
11.3%
Proportion of Dataset

Feb
Worst Month for Extremes

China
Most Events — 18 Records

02 | EXTREME EVENTS — By Country & Month





WHY THESE PATTERNS?

🌏 China & Saudi Arabia (18 each)

Both combine industrial pollution spikes (PM2.5 >100) with extreme UV summer exposure. Desert location amplifies heat events in Saudi Arabia.

🌴 Tropical Nations

Belize (16) and Indonesia (16) see extreme humidity consistently above 95% — driven by equatorial moisture and rainforest climate.

📅 February Worst Month

February peaks because Southern Hemisphere summer adds UV extremes while Northern Hemisphere still faces cold wind events — both extremes active simultaneously.

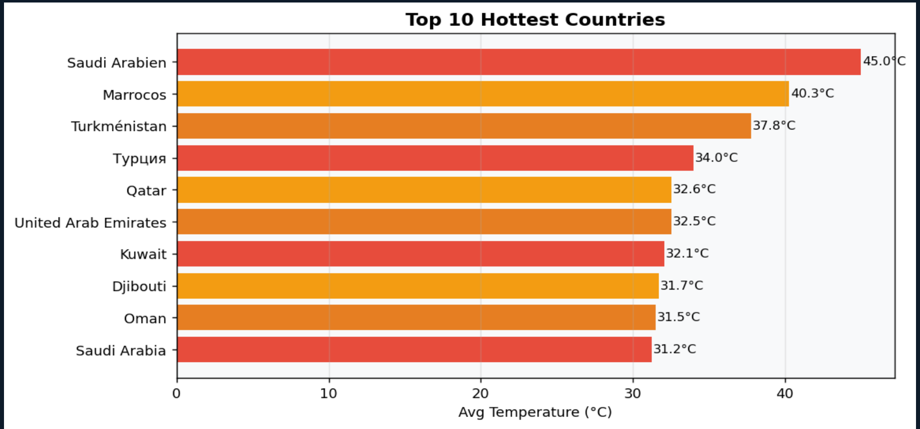
🌞 UV Dominates

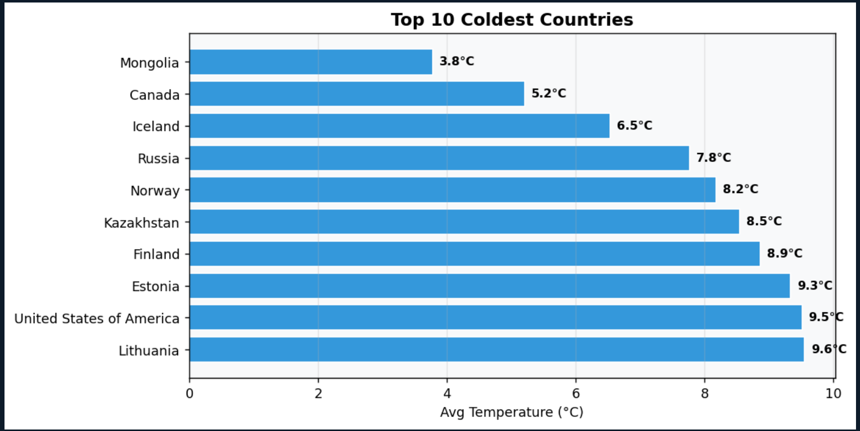
201 of 494 extreme records involve UV >10. This is driven by high-altitude nations (Nepal, Bolivia) and equatorial countries with minimal cloud cover.

🏭 Air Quality Crisis

118 records exceed PM2.5 100 μg/m³ — concentrated in South & East Asia (China, India, Kuwait). Cold winter inversions trap pollutants over cities.

03 | REGIONAL & COUNTRY COMPARISON — Who's Hottest, Coldest & Most Variable?





🌡️ TOP 10 HOTTEST

❄️ TOP 10 COLDEST

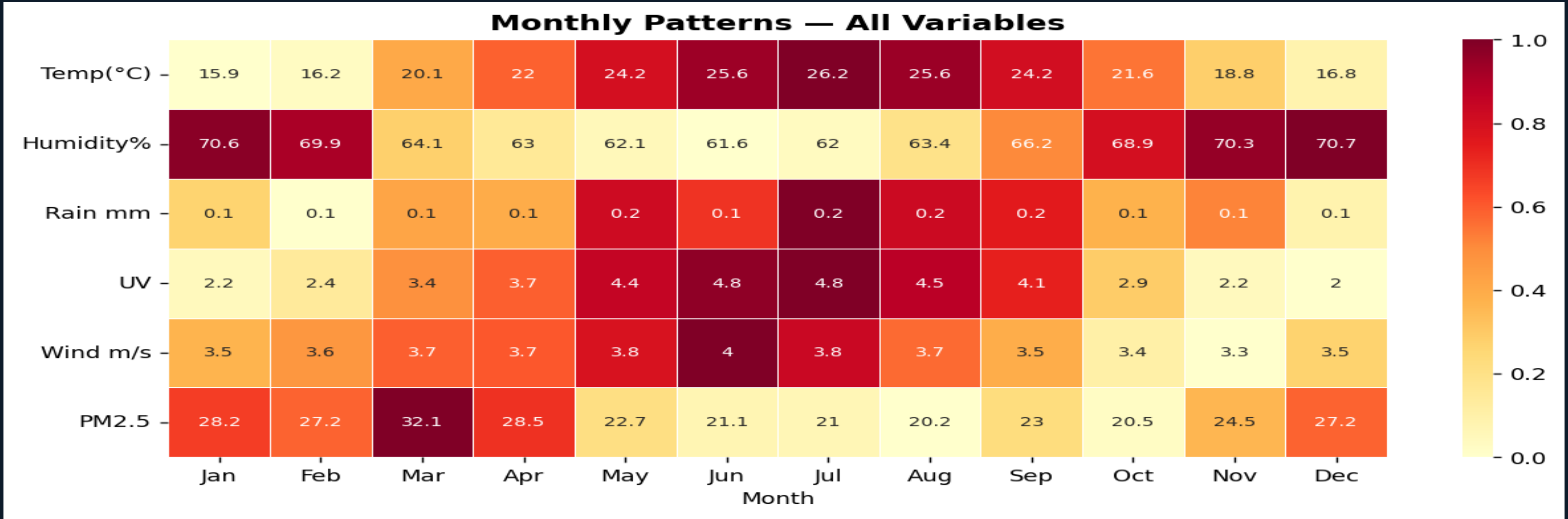
HOT COUNTRY INSIGHTS

Saudi Arabia leads at 45.0°C avg peak — driven by the Rub al Khali desert's continental climate. Morocco (40.3°C) and Turkmenistan (37.8°C) share arid Central Asian/North African conditions. Turkey (34°C) and Qatar (32.6°C) combine desert heat with Persian Gulf humidity. All top-5 countries lie within the subtropical high-pressure belt (15°N–40°N).

COLD COUNTRY INSIGHTS

Mongolia leads coldest at 3.8°C annual avg due to its extreme continental climate — no ocean to moderate temperatures. Canada (5.2°C) and Iceland (6.5°C) have large Arctic-influenced landmasses. Russia (7.8°C) despite its size stays cold due to Siberian highs. Norway (8.2°C) is warmed by the Gulf Stream but still registers cold due to its northern latitude (71°N).

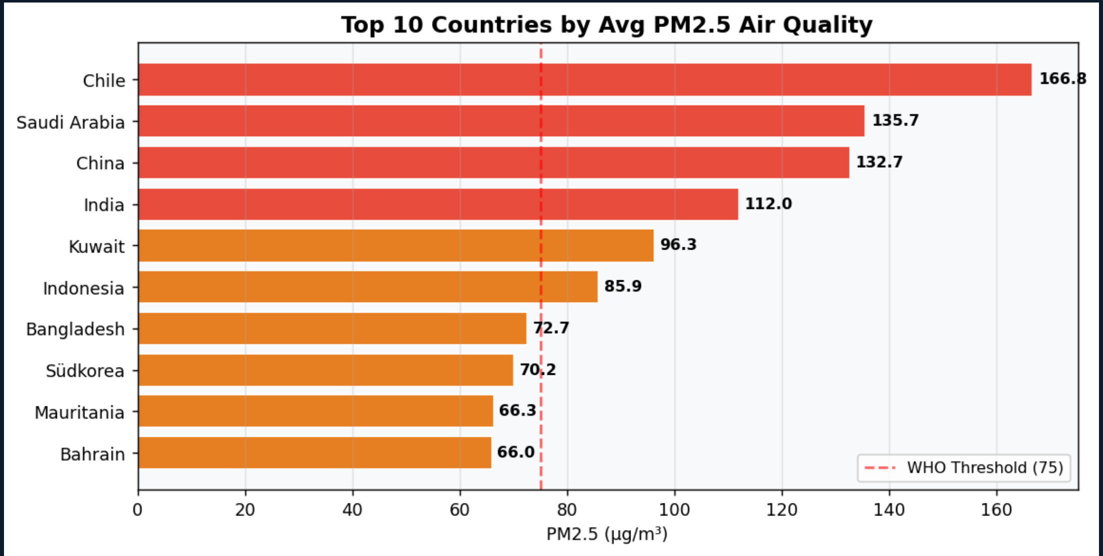
03 | MONTHLY TEMPERATURE HEATMAP — Seasonal Patterns by Country



READING THE HEATMAP

- Each cell shows the average temperature for a country (row) in a specific month (column). Darker red = hotter; deeper blue = colder.
- Saudi Arabia, Qatar, and Morocco maintain deep red throughout May–Oct, confirming consistent desert-level heat with limited seasonal variation.
- Mongolia shows the sharpest contrast — near-zero temperatures Jan–Mar shifting to moderate warmth Jun–Aug — widest range in the dataset.
- European nations (Norway, Iceland) show clear U-shaped patterns: cold winters, moderate summers. The gradient reveals the Atlantic's moderating effect.
- Countries near the equator (Belize, Indonesia) show flat horizontal patterns — minimal seasonal temperature variation year-round.

03 | AIR QUALITY ACROSS REGIONS — PM2.5 Pollution Analysis



AIR QUALITY FINDINGS

764.9	Max PM2.5 — Single Record
5 µg/m³	WHO Annual Guideline
Chile	Highest Avg — 166.8 µg/m³
118	Records Above 100 µg/m³

Top 5 Most Polluted:

Chile: 166.8 µg/m³

WHAT IS PM2.5 AND WHY DOES IT MATTER?

PM2.5 refers to fine particulate matter with diameter ≤2.5 micrometres — smaller than a human hair. These particles penetrate deep into lung tissue and enter the bloodstream, causing respiratory disease, cardiovascular damage and premature death. The WHO 2021 revised guideline is 5 µg/m³ annual mean. Cities in South Asia and the Middle East regularly exceed this by 20–150 times. High-PM2.5 months correlate strongly with cold winter inversions (China, India) and dust storm seasons (Saudi Arabia, Kuwait). PM2.5 and PM10 share a near-perfect correlation (r=0.97), confirming shared sources: traffic, industry and biomass burning.

04 | VISUALIZATION TYPES & RATIONALE — Choosing the Right Chart

🗺️ Choropleth Map

When to use: Comparing geographic distributions

Color-encodes a single metric across world countries. Reveals spatial clusters and hotspots at a glance.

Used for temperature, humidity, UV, PM2.5 across 211 countries. Instantly shows which regions are affected most.

🏳 Limitation: Cannot show change over time — best paired with a line chart for temporal trends.

📈 Line Chart

When to use: Showing trends over time

Plots a continuous variable against months. Multiple lines on one chart allow direct comparison of variables.

Reveals seasonality — temperature peaks in July, UV follows, humidity peaks in December. Direction of trend is immediately clear.

🏳 Best practice: Label peaks and troughs with data callouts. Use smooth curves for monthly climate data.

🔵 Scatterplot

When to use: Exploring relationships between two variables

Each point = one country-month observation. Colour by month adds a third dimension. Trend line reveals direction.

Shows that Temp and Humidity have a negative correlation (r=-0.34). Desert countries cluster at high-temp, low-humidity.

🏳 Watch for: Outliers (extreme PM2.5 cities) that skew the visual. Log scale may be needed for skewed axes.

🟥 Heatmap

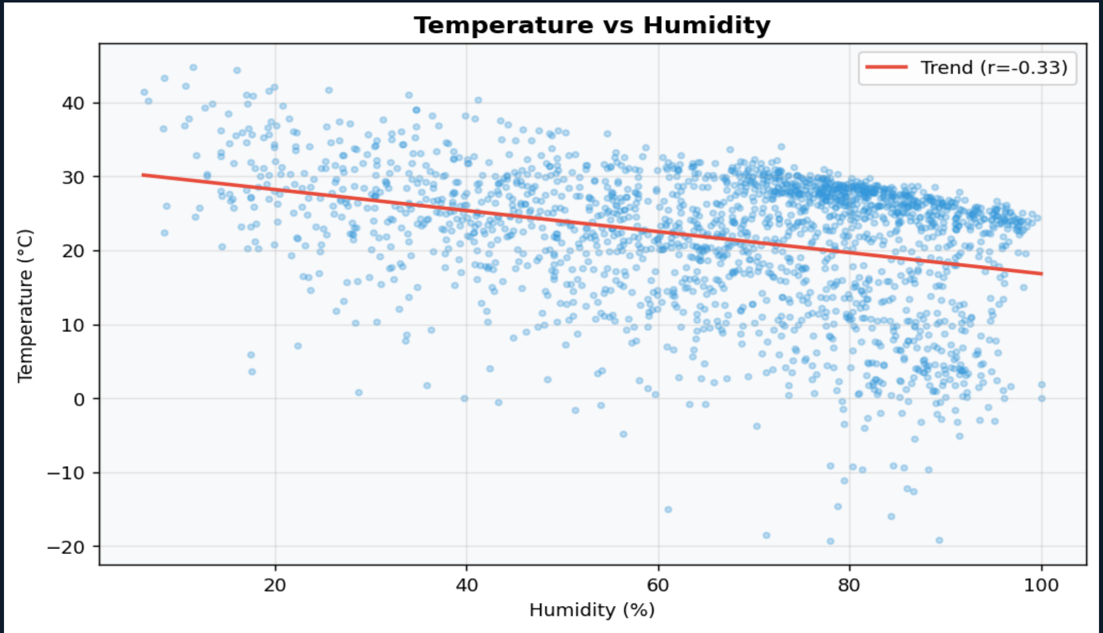
When to use: Many variables × many categories at once

Grid of coloured cells where row = country/variable, column = month. Colour intensity encodes magnitude.

Correlation heatmap reveals hidden relationships across 9×9 variable pairs. Seasonal heatmap shows country-by-month temperature patterns.

🏳 Key strength: Information density — shows 20 countries × 12 months = 240 data points in one view.

04 | SCATTERPLOT ANALYSIS — Temperature vs. Humidity



WHAT THE SCATTER REVEALS

Negative Correlation r = -0.34
As temperature rises, relative humidity tends to drop. Warm air holds more water vapour but relative humidity depends on actual vs. capacity — hot deserts are dry.

Tropical Cluster
Countries at 25–35°C, 80–100% humidity are equatorial rainforests (Indonesia, Belize). High temperature AND high humidity co-exist due to ocean moisture.

Desert Outliers
Saudi Arabia, Kuwait, UAE plot at extreme heat (35–45°C) and very low humidity (5–25%). Confirms continental desert climate physics.

Trend Line Slope: -0.16

For every 1°C rise in temperature, humidity drops approximately 0.16 percentage points on average across the dataset.

Arctic Cluster
Mongolia, Canada, Iceland cluster at 3–10°C with 50–70% humidity. Cold air can hold less water, explaining moderate relative humidity despite low temperatures.

INTERPRETING CLIMATE CLUSTERS

The scatterplot reveals four distinct climate clusters: (1) Hot-Dry — desert nations above 35°C with humidity below 30%; (2) Hot-Humid — tropical nations 25–35°C with humidity 80–100%; (3) Temperate — most mid-latitude nations clustering near the regression line; (4) Cold-Moderate — Arctic and sub-Arctic nations below 12°C. These clusters align precisely with the Köppen climate classification system.

KEY TAKEAWAYS — Summary of Findings from the Global Weather Analysis

01 Distributions

Weather variables are non-normal and heavily skewed. PM2.5 (max 764.9 μg/m³) shows severe outliers from polluted Asian cities — 153× the WHO limit. Temperature is the most normally distributed variable with a global mean of 21.5°C and std dev of 9.38°C.

▶ 21.5°C global mean temperature

02 Extreme Events

494 records (11.3%) qualify as extreme. UV events dominate (201 cases) driven by equatorial and high-altitude nations. February saw the most extremes due to simultaneous Northern cold and Southern UV peaks. China and Saudi Arabia lead with 18 events each.

▶ 11.3% of dataset = extreme

03 Seasonal Trends

Global temperatures peak in July (26.2°C) and trough in January (15.9°C) — a 10°C seasonal swing. UV and temperature move together (r=+0.51). Humidity peaks in December due to Southern Hemisphere monsoon and ocean effects.

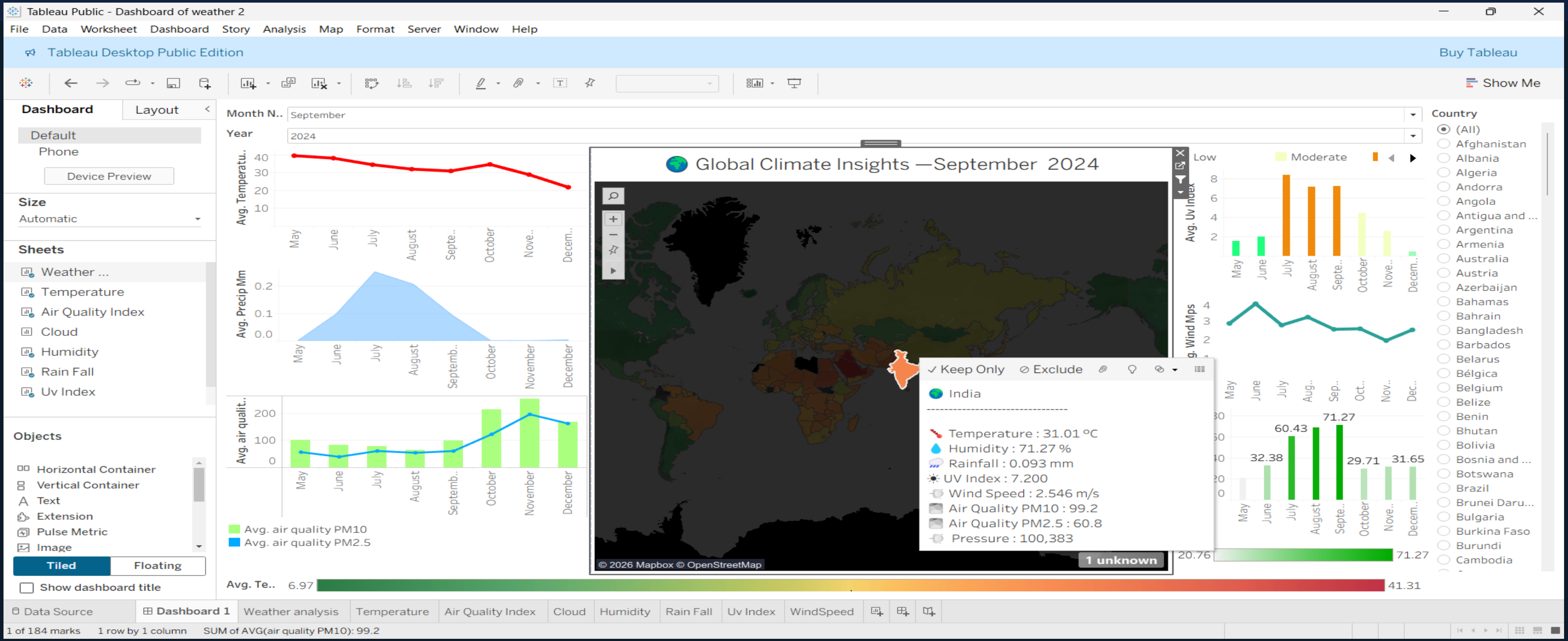
▶ +10°C seasonal temperature swing

04 Regional Insights

Middle East and North Africa are consistently the hottest regions. Asia holds the worst air quality driven by industrial pollution. Europe is the most climatically stable continent. Mongolia (-20.5°C) and Saudi Arabia (+45°C) define the dataset's temperature extremes.

▶ 45°C max vs -20.5°C min globally

05 | INTERACTIVE DASHBOARD — Global Climate Insights



🗺️ Choropleth World Map

India highlighted in Sep 2024. Colour gradient encodes Avg. Temp (6.97°C–41.31°C). Users can select any country from the right panel.

📈 Temperature Trend Line

Selected region shows avg temp declining from ~40°C (May) to ~23°C (Dec). Confirms post-monsoon cooling in South Asia.

🌧️ Precipitation Chart

Peak rainfall in Aug–Sep (~0.25mm) maps to Indian monsoon season. Sharp post-October decline confirms seasonal pattern.

🏭 Air Quality (PM10 & PM2.5)

PM10 spikes dramatically in Oct–Nov (>200 µg/m³). PM2.5 tracks below PM10. Post-monsoon stubble burning drives the surge.

THANK YOU

Global Weather Analysis — A Data-Driven Exploration of Climate Patterns

4,385	211	9	7
Records Analysed	Countries Covered	Weather Variables	Extreme Event Types

This analysis demonstrates that climate data visualization can transform raw weather numbers into compelling stories about human geography, environmental health, and seasonal rhythms. From Mongolia's -20.5°C winters to Saudi Arabia's 45°C summers — data brings the world's climate into sharp focus.

Presented by: Durga Prasad Singothu · Mentor: Shanmukha Priya · Group 2 · Infosys Springboard · Milestone 2